
COVERAGE CERTIFICATES UNDER NONIDENTIFIABLE LABEL SHIFT

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Paper under double-blind review

ABSTRACT

Label shift changes class proportions while preserving class-conditional feature distributions. Estimating every target class proportion can be impossible when a fixed summary merges classes, yet the coverage of a particular prediction set can remain identifiable. We develop finite-sample coverage certificates that exploit this distinction. For an independently fixed finite library of prediction sets, class-stratified source samples and unlabeled target summaries define a polytope of plausible target priors. Maximizing estimated miscoverage over this polytope yields simultaneous upper confidence bounds without a matrix inverse. Their excess over true risk is bounded by source estimation error and a modulus measuring risk variation between nearly indistinguishable priors. A linear-program dual separates summary sensitivity from residual risk variation. We connect the certificate to the classical row-space criterion for functional identification and derive a two-point lower bound for uniformly honest risk intervals. Explicit classification models exhibit informative singular cases and a finite-sample obstruction near coincident class-conditionals. Repeated categorical experiments illustrate shrinking uncertainty for aligned risks and persistent ambiguity for nonaligned risks. The resulting analysis separates the information needed to certify target marginal coverage from the stronger information needed to recover the entire target prior.

1 INTRODUCTION

A prediction set expresses uncertainty by retaining several plausible class labels. Its usefulness depends on controlling the probability that the true label is excluded, together with the size of the retained set. When class proportions change between calibration and deployment, an error rate measured on the source population need not describe the target population. Unlabeled target observations can reveal this change, but only through the information contained in their features or in the summaries retained for calibration.

Label shift provides a structured formulation of this problem: the conditional feature law within each class remains fixed, while the class probabilities vary. Estimation methods based on a classifier’s confusion matrix recover the target proportions when the observation matrix is invertible (Lipton et al., 2018). Coverage methods can then incorporate class importance weights, either through weighted conformal inference or through confidence bounds for those weights (Podkopaev & Ramdas, 2021; Si et al., 2024). This approach raises a distinct question when two classes have the same summary distribution. Full recovery of the target proportions is then impossible, but does certification of a particular prediction set also become impossible?

The answer depends on which errors the prediction set makes. Suppose a summary cannot distinguish labels 1 and 2 but distinguishes both from label 3. A constant set containing labels 1 and 2 misses exactly label 3, whose prevalence remains observable. A constant set containing only label 1 instead requires separating labels 1 and 2 to determine its risk. Both rules face the same singular observation matrix. Their inferential requirements differ because their class-conditional error vectors differ. The relevant object is therefore the variation of

a risk functional over target priors that produce the same, or nearly the same, observed summaries.

We give a finite-sample analysis organized around that object. First, independent class-stratified source data estimate both the summary matrix and the class-conditional error vectors. Unlabeled target data estimate the summary distribution. Simultaneous concentration produces a polytope containing the true target prior with prescribed confidence. A linear program maximizes each estimated error functional over that polytope. The resulting upper bounds remain valid under singularity and under selection of a member of the fixed finite library after calibration. The quantitative error bound separates sampling error from the ambiguity of the particular risk functional.

Second, we state and prove the exact identified risk interval on each population fiber. Its collapse on every fiber is equivalent to row-space membership of the error vector, an elementary functional-identification criterion. This characterization supplies both informative singular examples and pairs of observationally indistinguishable priors whose risks differ. Third, a two-point argument extends the obstruction to finite samples: uniformly honest risk intervals must retain positive length when two priors have close summary laws. A binary construction makes the dependence on class separation explicit.

The analysis combines established ideas from upper-confidence-bound risk calibration, convex functional estimation and statistical testing (Bates et al., 2021; Juditsky & Nemirovski, 2009). Its specific formulation propagates uncertainty in the sampled observation matrix and sampled error vectors while retaining the geometry of a fixed prediction-set library. The contribution is an explicit certificate and a connected account of its information boundary; the LP principle, row-space algebra and two-point testing method are established tools. Synthetic experiments examine the resulting conservatism and the distinction between shrinking sampling uncertainty and persistent identification ambiguity. All coverage assertions concern target marginal risk under exact label shift and a future observation independent of calibration.

2 RELATED WORK

Prediction sets and risk control. Set-valued classification separates error control from ambiguity, commonly measured by expected set size. Oracle ambiguity minimization and its statistical estimation are developed by Sadinle et al. (2018). Adaptive classification scores and regularization improve the behavior of conformal prediction sets (Romano et al., 2020; Angelopoulos et al., 2020); quantile-based constructions also address uneven coverage across easy and difficult observations (Cauchois et al., 2021). These approaches motivate informative libraries of candidate sets. The results below certify a fixed library and leave its approximation to a population-optimal set unresolved.

Upper confidence bounds provide a general mechanism for controlling the risk of a selected set. Theorem 1 of Bates et al. (2021) calibrates nested monotone families using risk bounds; their discussion also treats uniform control for nonmonotone families. The finite-library selection step below uses that established principle. Conformal risk control instead targets the expected value of a monotone loss (Angelopoulos et al., 2022). The two confidence statements differ: a high-probability bound on conditional target risk contains a separate calibration-failure parameter, while an expected-risk guarantee averages over calibration.

Label-shift information and uncertainty. Moment matching through classifier outputs yields consistent prior estimates and finite-sample error bounds under an invertible confusion matrix (Lipton et al., 2018). Regularized label-shift learning incorporates uncertainty in estimated weights (Azizzadenesheli et al., 2019); the unified analysis of moment matching and maximum likelihood identifies the roles of calibration and invertibility (Garg et al., 2020). Fisher-consistency analysis provides another criterion for selecting prior estimators (Tasche, 2017). Under separation conditions, ratio estimators admit upper and lower rate results (Vaz et al., 2019), while nonparametric classification under label shift has a broader minimax analysis (Maity et al., 2022). These results concern prior estimation or

classification error; the functional treated below can be identified even when the whole prior is not.

For prediction sets, Theorem 2 of Podkopaev & Ramdas (2021) establishes weighted conformal coverage using true likelihood ratios, and their Theorem 4 quantifies calibration-error transfer when weights are estimated. Si et al. (2024) construct PAC sets with confidence intervals for importance weights: Lemma 3.1 propagates uncertainty through interval Gaussian elimination, and Theorem 3.2 supplies the coverage guarantee under the stated label-shift and invertibility assumptions. Their Appendix D relaxes intermediate algorithmic positivity conditions. The certificate below instead optimizes directly over plausible simplex-valued priors. This preserves validity for singular summary matrices, at the cost of retaining any risk ambiguity that the summaries cannot resolve.

Convex functional inference and robust sets. Estimating a linear functional through a known affine observation map over a compact convex parameter set is a classical inverse-inference problem. Juditsky & Nemirovski (2009) construct convex-programming estimators and relate confidence risk to Hellinger affinity and functional variation; their framework includes positive discrete observation families. That work gives a deeper minimax treatment of the known-operator problem. The sampled-operator certificate below uses elementary concentration to account for uncertainty in both the map and the risk vector. Its upper and lower bounds use different moduli and are not asserted to match universally.

Credal-set optimization also yields robust prediction sets. Proposition 4.1 of Javanmardi et al. (2025) formulates minimum-size randomized sets using a supplied instance-level credal polytope. The prior polytope below is estimated from source strata and unlabeled target summaries and supports marginal coverage for a fixed library. The uncertainty object and statistical construction differ, while the robust linear-programming principle overlaps. These distinctions locate the analysis within existing convex risk-control methods.

Other departures from exchangeability. Weighted conformal prediction under covariate shift uses likelihood-ratio information for a different invariant conditional law (Tibshirani et al., 2019). Practical adaptation of prediction sets from unlabeled shifted data has also been studied through score adjustment (Kasa et al., 2025). Online procedures address evolving distributions (Gibbs & Candès, 2024), and concentration with decoupling can accommodate nonexchangeable observations (Oliveira et al., 2024). Label noise changes which labels are observed (Einbinder et al., 2024); generalized label shift can additionally vary latent nuisance distributions (Masserano et al., 2024). These settings identify substantive extensions beyond the fixed class-conditional, independent-sample model used in the following results.

3 LABEL SHIFT AND RISK FUNCTIONALS

Let the label set be $[K] = \{1, \dots, K\}$ and let $\mathcal{X}$ be a measurable feature space. The positive integers K, d, L are fixed. A measurable summary $g : \mathcal{X} \rightarrow [d]$ and measurable prediction sets $C_1, \dots, C_L : \mathcal{X} \rightarrow 2^{[K]}$ are fixed independently of calibration. The library includes the full-label rule $C_{\text{all}}(x) = [K]$. Training may precede calibration; all assertions can be read conditionally on independently trained functions.

Sampling model. For every class y , the source observations $X_{y1}, \dots, X_{ym_y}$ are independent draws from a conditional law P_y , with $m_y \geq 1$. Different source strata are independent. Independently, $n \geq 1$ target observations are drawn from $\sum_y q_y P_y$, where the unknown target prior belongs to

$$\Delta_K = \{v \in \mathbb{R}^K : v_y \geq 0, \mathbf{1}^\top v = 1\}.$$

Only their summaries $Z_j = g(X_j)$ are needed for calibration. A future labeled target observation (X, Y) is independent of all calibration data and has prior q and conditional law P_y . Equality of these class-conditional laws across source and target is the exact label-shift assumption.

Define the column-stochastic summary matrix and the class-conditional error vector of rule ℓ by

$$M_{iy} = P_y\{g(X) = i\}, \quad r_{\ell y} = P_y\{y \notin C_\ell(X)\}. \quad (1)$$

The observable summary distribution and target miscoverage risk satisfy

$$p = Mq, \quad R_\ell(q) = r_\ell^\top q. \quad (2)$$

No calibration property of a classifier's probability outputs is required for these identities. The summary can be a predicted label, a bin, or another predetermined categorical statistic. Coarsening it changes M and can remove information about q .

Write $\widehat{M}, \widehat{r}_\ell$ for the empirical averages of the indicators in (1) within each source stratum, and $\widehat{p}_i = n^{-1} \sum_j \mathbf{1}\{Z_j = i\}$. For a confidence parameter $\delta \in (0, 1)$, set

$$\tau = \sqrt{\frac{\log(4d/\delta)}{2n}}, \quad e_y = \sqrt{\frac{\log(4K(d+L)/\delta)}{2m_y}}, \quad e = \max_y e_y, \quad \rho = \tau + e. \quad (3)$$

The feasible prior polytope is

$$\widehat{Q} = \{v \in \Delta_K : \|\widehat{M}v - \widehat{p}\|_\infty \leq \rho\}. \quad (4)$$

All vector infinity norms are maximum absolute coordinates. Bounds on matrix entries below mean the largest absolute entry, rather than an induced matrix norm.

For a risk vector r , define its ambiguity modulus for $t \geq 0$ by

$$\Omega_r(t) = \sup_{\substack{u, v \in \Delta_K \\ \|M(u-v)\|_\infty \leq t}} |r^\top(u-v)|. \quad (5)$$

Use $\Omega_\ell = \Omega_{r_\ell}$. This modulus is a population quantity used to interpret certificate efficiency; computing the certificate itself requires only empirical quantities. At $t = 0$ it measures risk variation between exactly indistinguishable priors. The simplex constraint matters: it retains nonnegativity and normalization even when an unrestricted linear system has many solutions. Because risks lie in $[0, 1]$, $\Omega_r(t) \leq 1$ for the error vectors under consideration.

4 FINITE-SAMPLE COVERAGE CERTIFICATES

A plausible-prior region can be useful without selecting a single prior estimate. To certify a rule, it suffices to upper-bound its risk at every prior in that region. When $\widehat{Q} \neq \emptyset$, define

$$V_\ell = \max_{v \in \widehat{Q}} \widehat{r}_\ell^\top v, \quad U_\ell = \min\{1, e + V_\ell\}. \quad (6)$$

If $\widehat{Q} = \emptyset$, set $U_\ell = 1$. The full-label rule may always receive the exact certificate $U_{\text{all}} = 0$. Both conventions define the procedure for every possible calibration sample.

Theorem 1 (Simultaneous risk certificates). *Under the sampling model of Section 3, with probability at least $1 - \delta$ over calibration, $q \in \widehat{Q}$ and, simultaneously for every ℓ ,*

$$R_\ell(q) \leq U_\ell \leq \min\{1, R_\ell(q) + 2e + \Omega_\ell(2\rho)\}. \quad (7)$$

For every nonempty $\widehat{Q}$, the optimization in (6) has the exact dual representation

$$V_\ell = \inf_{a \in \mathbb{R}^d} \left\{ a^\top \widehat{p} + \rho \|a\|_1 + \max_{y \in [K]} (\widehat{r}_{\ell y} - (\widehat{M}^\top a)_y) \right\}. \quad (8)$$

Furthermore, with $\text{osc}(b) = \max_y b_y - \min_y b_y$,

$$\Omega_\ell(t) \leq \inf_{a \in \mathbb{R}^d} \{t \|a\|_1 + \text{osc}(r_\ell - M^\top a)\}, \quad t \geq 0. \quad (9)$$

For $\alpha \in [0, 1]$, any measurable choice of an index satisfying $U_\ell \leq \alpha$, with the full-label rule as fallback, has target marginal coverage at least $1 - \alpha$ on the same calibration event.

The complete derivation is in Appendix A.1.

Why the empirical polytope is enough. Three steps explain the guarantee. First, a simultaneous concentration event bounds target-coordinate error by τ and source-coordinate error by e . Dependence between different indicators on one source example does not prevent this union bound, since each empirical average uses independent observations. On that event, simplex weights imply $\|(\widehat{M} - M)q\|_\infty \leq e$, placing q in $\widehat{Q}$.

Second, every feasible v is close to q in its observable image:

$$\begin{aligned} \|M(v - q)\|_\infty &\leq \|(M - \widehat{M})v\|_\infty + \|\widehat{M}v - \widehat{p}\|_\infty + \|\widehat{p} - Mq\|_\infty \\ &\leq e + \rho + \tau = 2\rho. \end{aligned} \tag{10}$$

This step replaces control of $\|v - q\|$ by control of the quantity the summaries actually identify. It remains valid when the null space of M is nontrivial. Third, the modulus bounds risk variation over feasible images, while the empirical risk vector contributes an additional e on each side of the comparison. The total source contribution is $2e$ in (7).

Dual interpretation. The vector a in (8) assigns a coefficient to each summary category. The first term estimates the resulting target expectation, the second penalizes sensitivity to the uncertainty radius, and the last bounds residual error across classes. If a risk vector is accurately represented by $M^\top a$, its functional ambiguity can be small even when M is singular. Conversely, a large residual oscillation prevents a small bound from following solely from summary concentration. Equation (9) permits approximate alignment: exact row-space membership is sufficient but not required for a useful finite-resolution bound.

The primal and dual are finite-dimensional linear programs after introducing variables for absolute values and the coordinatewise maximum. Strong duality needs feasibility and boundedness, supplied by a nonempty subset of the simplex; strict feasibility and invertibility are unnecessary. One feasible polytope serves all rules, with only the objective changing. This exposes which aspects of the computation are shared across the library.

Selection and confidence. Simultaneity permits choosing a certified rule using calibration data, for example the least-index certified member under a predetermined ordering. An independently trained ordering can favor smaller or otherwise preferred sets. Minimum expected size is a separate optimization objective. The factor L in the source radius accounts for testing the finite family; adding calibration-dependent rules changes the statistical problem.

Conditional on a calibration realization in the common event, the selected rule is fixed and its future target miscoverage is at most α . Averaging over all calibration realizations gives the bound $\alpha + \delta$, since risk is bounded by one on the exceptional event. A PAC guarantee and an unconditional nominal- α guarantee are therefore distinct. This distinction motivates risk-controlling sets (Bates et al., 2021) and PAC label-shift sets (Si et al., 2024).

The radius e is governed by the least-sampled source class. Increasing the target sample alone cannot remove source uncertainty. Moreover, a small radius need not yield a small certificate if the error functional varies along a population fiber. The next result identifies when that residual ambiguity disappears.

5 IDENTIFICATION ON SINGULAR MIXTURE FIBERS

The certificate contains uncertainty about the observed distribution and uncertainty that persists even if that distribution is known exactly. The latter can be examined without estimating a matrix. Fix a column-stochastic M and $r \in [0, 1]^K$. For $p \in M\Delta_K$, write

$$F(p) = \{v \in \Delta_K : Mv = p\}.$$

For statistical lower bounds, the complete observation is $(Z_1, \dots, Z_n, W)$ with law

$$\mathbb{P}_q = (Mq)^{\otimes n} \otimes H. \tag{11}$$

The summaries are independent and identically distributed. Auxiliary data W are independent of them and have a common law H not depending on q ; any random seed is included

in W . The fixed matrix and conditional laws may be supplied exactly. Independent source calibration is admissible auxiliary information. An interval I is uniformly honest at level $1 - \delta$ if $\mathbb{P}_q\{r^\top q \in I\} \geq 1 - \delta$ for every $q \in \Delta_K$.

Proposition 2 (Risk identification and indistinguishability). *In the finite-label setting of Section 3, the sharp population risk region at p is*

$$\mathcal{R}(p) = \left[\min_{v \in F(p)} r^\top v, \max_{v \in F(p)} r^\top v \right]. \quad (12)$$

It is a singleton for every compatible p if and only if $r \in \text{row}(M)$. If r is not row-aligned, there are interior priors q_0, q_1 with $Mq_0 = Mq_1$ and different risks. Under (11), every uniformly honest interval satisfies

$$\mathbb{P}_{q_0}\{|I| \geq |r^\top(q_1 - q_0)|\} \geq 1 - 2\delta$$

for any such pair. If r is row-aligned, then

$$\Omega_r(t) \leq t \inf_{a: M^\top a = r} \|a\|_1, \quad t \geq 0.$$

Appendix A.2 proves all assertions, including the simplex and boundary cases.

Compactness and convexity give (12): the endpoints are attained, and mixtures of endpoint priors attain every intermediate risk. Row alignment gives $r^\top v = a^\top p$, constant on a fiber. For necessity, a nonaligned vector has a nonzero component along $\ker M$; a small perturbation of the uniform prior in that direction preserves positivity, normalization and the observed distribution while changing risk. Finally, (11) makes all available information identical at the two priors. Uniform interval coverage forces inclusion of both risks with probability at least $1 - 2\delta$.

The quantifier *for every compatible p* is essential. A boundary fiber can be a singleton even when r is not globally row-aligned. The criterion is an elementary instance of linear-functional identification, consistent with the convex inverse-inference viewpoint of Juditsky & Nemirovski (2009). Its role is to identify the information required by a particular classification loss.

A realized singular classification problem. Let $\mathcal{X} = [2]$, $g(x) = x$, and

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Labels 1 and 2 generate summary 1, and label 3 generates summary 2. For $C(x) = \{1, 2\}$, the error vector is $(0, 0, 1)^\top = M^\top(0, 1)^\top$. Its risk is $q_3 = p_2$ and its set size is two. By comparison, $C(x) = \{1\}$ has error vector $(0, 1, 1)^\top$, and at $p = (3/4, 1/4)$ its identified interval is $[1/4, 1]$. The interior priors $(1/2, 1/4, 1/4)$ and $(1/4, 1/2, 1/4)$ give risks $1/2$ and $3/4$ for the second rule. The singularity is identical; the distinction comes entirely from the error vector. Requiring a reliable full-prior estimate can thus impose an unnecessary obstacle to coverage certification.

6 CONFIDENCE LIMITS NEAR NONIDENTIFIABILITY

Even when the columns of M differ, a small separation can make summary distributions difficult to distinguish from finite samples. We quantify this obstruction in (11), retaining the finite-label and positive-sample-size conditions of Section 3. For categorical laws, use

$$h^2(p_0, p_1) = \sum_{i=1}^d (\sqrt{p_{0i}} - \sqrt{p_{1i}})^2.$$

The absence of a factor $1/2$ fixes the constants below.

Theorem 3 (Two-point confidence-length bound). *Let $\delta \in (0, 1/4)$ and let $q_0, q_1 \in \Delta_K$ satisfy $h^2(Mq_0, Mq_1) \leq 1/(16n)$. Under (11), every uniformly honest interval for $r^\top q$ satisfies*

$$\mathbb{P}_{q_0}\{|I| \geq |r^\top(q_1 - q_0)|\} \geq \frac{3}{4} - 2\delta. \quad (13)$$

Consequently,

$$\sup_q \mathbb{E}_q |I| \geq (\tfrac{3}{4} - 2\delta) \sup_{\substack{q_0, q_1 \in \Delta_K \\ h^2(M_{q_0}, M_{q_1}) \leq 1/(16n)}} |r^\top (q_1 - q_0)|.$$

For $r = (0, 1)^\top$ and

$$M_\kappa = \begin{pmatrix} 1/2 + \kappa & 1/2 - \kappa \\ 1/2 - \kappa & 1/2 + \kappa \end{pmatrix}, \quad 0 \leq \kappa \leq 1/4,$$

the last supremum is at least $\min\{1/2, (16\kappa\sqrt{n})^{-1}\}$, with the second term interpreted as infinity when $\kappa = 0$. This risk vector is realized by the constant prediction set $\{1\}$.

Appendix A.3 gives the complete probability and constant calculations.

The proof follows three steps. Cauchy–Schwarz bounds total variation by Hellinger distance, and independence makes the affinity of product laws multiply. Hence

$$\text{TV}(\mathbb{P}_{q_0}, \mathbb{P}_{q_1}) \leq \sqrt{n h^2(M_{q_0}, M_{q_1})} \leq \tfrac{1}{4}. \quad (14)$$

The common independent law H leaves this comparison unchanged. Next, transfer the event that I contains $r^\top q_1$ to the law at q_0 , losing at most $1/4$ probability, and intersect it with coverage of $r^\top q_0$. Finally, choose $q_0 = (1/2 + t/2, 1/2 - t/2)$ and q_1 with swapped coordinates, where $t = \min\{1/2, (16\kappa\sqrt{n})^{-1}\}$ for $\kappa > 0$. The risks differ by t , while

$$h^2(M_{\kappa q_0}, M_{\kappa q_1}) = 2 - 2\sqrt{1 - 4\kappa^2 t^2} \leq 8\kappa^2 t^2 \leq \frac{1}{32n}.$$

The pair lies within the testing radius. At $\kappa = 0$, take $t = 1/2$ directly.

The obstruction concerns two-sided uniformly honest intervals in the specified summary experiment. More informative features can distinguish priors that a coarse summary merges. The theorem bounds a worst-case expected length, rather than the length at every prior. The construction illustrates a transition from an inverse-separation scale to nonvanishing ambiguity, without matching the dimensional or logarithmic factors of the certificate. Hellinger-modulus functional inference is established in Juditsky & Nemirovski (2009), and label-shift minimax analysis under separation is developed by Vaz et al. (2019). The construction supplies explicit confidence-interval constants and an actual classification realization for this comparison.

7 SYNTHETIC EXPERIMENTS

The simulations separate validity from informativeness. All data are generated from categorical models satisfying exact label shift, with independent source strata and target samples. We use $\delta = .05$ and 500 repetitions per regime. Source class counts equal the target count $n \in \{100, 400, 1600, 6400, 25600\}$. Linear programs use the radii in (3). For plotting, the risk interval has lower endpoint $\max\{0, \min_{\hat{Q}} \hat{r}^\top v - e\}$ and upper endpoint (6); it is $[0, 1]$ if the polytope is empty. Appendix B gives the models and numerical conventions.

Singular risks. For the three-class model in Section 5, both tested priors have summary law $(.75, .25)$. Figure 1 contrasts aligned and nonaligned error vectors with identical sampling budgets. The aligned mean interval width decreases from 0.785 at $n = 100$ to 0.0670 at $n = 25600$. The nonaligned width decreases from 1 to 0.784, remaining close to its population value .75. This persistence is an identification phenomenon. At nominal risk $\alpha = .35$, choosing $\{1, 2\}$ when certified and the full-label set otherwise gives mean size three through $n = 1600$ and size two at $n = 6400$ and 25600 in every repetition. The smaller rule’s true target risk is .25.

Sampling the operator. A separate nonsingular model uses $M = ((.8, .2), (.2, .8))$ in row notation and target prior $(.15, .85)$. Independent source counts estimate its entries. For the singleton rule $C(x) = \{g(x)\}$, whose true risk is .2, the mean upper certificate decreases from .401 to .212 across the sample-size grid. No simultaneous upper-certificate failure occurs in 500 repetitions at any sample size, in either model. For each individual regime, zero failures yield a one-sided exact 95% upper bound of 0.00598 on the failure probability. These per-regime bounds are not simultaneous over the entire experiment.

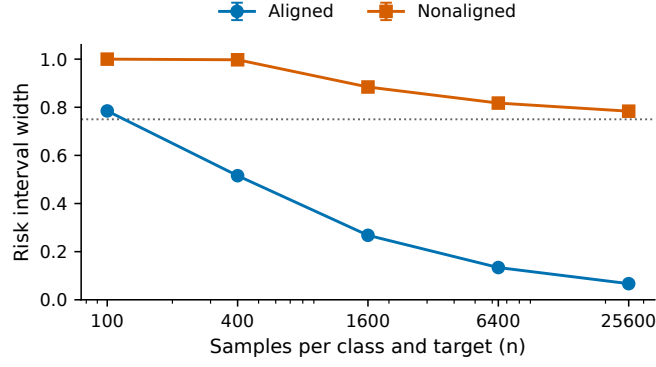


Figure 1: Risk-specific ambiguity at a singular mixture matrix $M = [e_1, e_1, e_2]$. We use $p = (.75, .25)$, $m_y = n$, $\delta = .05$, and 500 independent repetitions per sample size. Curves show mean widths of the empirical risk intervals for the constant sets $\{1, 2\}$ (aligned) and $\{1\}$ (nonaligned); error bars are ± 1.96 Monte Carlo standard errors and may be smaller than markers. The dotted line is the nonaligned population width .75.

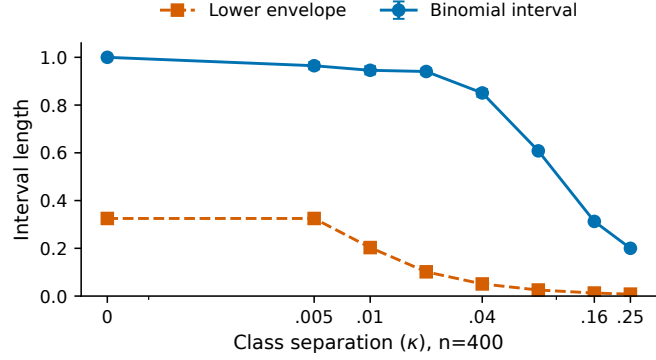


Figure 2: Confidence length near coincident class-conditionals. In the binary M_κ model, $n = 400$, $\delta = .05$, and $q = (1/2, 1/2)$, the blue curve is the mean length of transformed exact binomial intervals over 500 repetitions; bars are ± 1.96 Monte Carlo standard errors. The dashed curve is the deterministic worst-case lower envelope from Theorem 3. It bounds a supremum over priors, rather than the pointwise blue curve. The horizontal scale is linear up to .005 and logarithmic beyond.

Weak identification. For the binary family in Theorem 3, supply M_κ exactly, set $n = 400$, and sample at $q = (1/2, 1/2)$. An exact binomial interval for the first summary probability is mapped back to risk q_2 and intersected with $[0, 1]$. Figure 2 shows mean lengths near one for small separation, falling to .200 at $\kappa = .25$. The lower envelope concerns the supremum over priors, whereas the empirical curve uses the center prior. Their gap therefore measures neither constant optimality nor a theorem violation. The comparison illustrates deterioration near indistinguishable class-conditionals. These experiments evaluate finite categorical mechanisms and do not establish performance on deployed classification systems.

8 LIMITATIONS

The certificate requires equality of class-conditional laws. A change in those laws can alter both the summary matrix and the error vectors, so increasing the calibration sample does not correct this misspecification. The summary and finite library must also be fixed independently of calibration; adaptation outside that library requires additional complexity

control. Class-stratified sampling makes every source class available, while the radius is governed by its least-sampled stratum.

The bounds concern marginal target risk; conditional-on-feature coverage and minimum expected size require additional analysis. The ambiguity modulus measures information retained by the summary, and richer features can have smaller unidentified fibers. The two-point bound applies to the complete product experiment and uniformly honest two-sided intervals. Its constants and Hellinger geometry do not establish general optimality of the Hoeffding-polytope certificate. Numerical evidence is restricted to small categorical models and the tested grid.

9 CONCLUSION

Target coverage can depend on less information than full recovery of target class proportions. A simultaneous confidence polytope turns that observation into a finite-sample certificate, while its risk image separates estimation uncertainty from population ambiguity. Row alignment identifies informative singular cases, and a two-point construction quantifies the obstruction near indistinguishability.

Two rules exposed to the same summary matrix can therefore have different inferential limits. A precise direction is to choose summaries and prediction-set libraries jointly, using independent training data, so that retained risk functionals have small ambiguity while the resulting sets remain informative.

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APPENDIX

A PROOFS OF THE MAIN RESULTS

A.1 PROOF OF THEOREM 1

Proof. We prove Theorem 1. We first construct one calibration event and then establish every certificate on that event. The remaining dual and alignment identities are deterministic. Throughout the proof, the summary and library are fixed independently of calibration, as stipulated in Section 3.

A concentration lemma. Let B be Bernoulli with mean b . For $0 < b < 1$, define $f(t) = \log \mathbb{E} \exp(t(B - b))$. Differentiation identifies $f''(t)$ as the variance of a Bernoulli variable under exponential tilting. Every Bernoulli variance is at most $1/4$, so $f''(t) \leq 1/4$ for all real t . Since $f(0) = f'(0) = 0$, integration, for positive and negative t , gives

$$\mathbb{E} e^{t(B-b)} \leq e^{t^2/8}.$$

At $b = 0$ or $b = 1$, $B - b = 0$ almost surely and the same inequality holds directly. If $B_1, \dots, B_m$ are independent with this common mean, independence and Markov's inequality imply, for $t > 0$,

$$\mathbb{P} \left\{ \frac{1}{m} \sum_{j=1}^m B_j - b \geq s \right\} \leq \exp(-tms + mt^2/8).$$

Taking $t = 4s$ gives $\exp(-2ms^2)$ for $s > 0$. Applying the same calculation to the negative deviation yields

$$\mathbb{P}\{|\bar{B} - b| > s\} \leq 2e^{-2ms^2}. \quad (15)$$

Only independence within an individual empirical average enters this calculation.

The simultaneous event. Apply (15) to the d target indicators, each with radius τ . For each coordinate, the failure probability is at most $2e^{-2n\tau^2} = \delta/(2d)$. For source class y , apply it to the d summary indicators and the L error indicators, each with radius e_y . Each individual failure probability is at most $\delta/[2K(d+L)]$. The target union costs at most $\delta/2$, and the union over every source function and stratum costs at most $\delta/2$.

Consequently there is an event E with $\mathbb{P}(E) \geq 1 - \delta$ such that

$$\|\widehat{p} - p\|_\infty \leq \tau, \quad |\widehat{M}_{iy} - M_{iy}| \leq e_y \leq e, \quad |\widehat{r}_{\ell y} - r_{\ell y}| \leq e_y \leq e \quad (16)$$

for all indices. Different functions evaluated on the same source observation can be dependent: the union bound has not assumed independence between them. The class-stratified iid assumptions justify each source average. Independent target draws justify the target averages. Exact class-conditional invariance gives $p = Mq$ and $R_\ell(q) = r_\ell^\top q$ by the law of total probability.

Feasibility and validity. Work on E . For any $v \in \Delta_K$, nonnegative coordinates summing to one give

$$\|(\widehat{M} - M)v\|_\infty \leq \max_i \sum_y v_y |\widehat{M}_{iy} - M_{iy}| \leq e.$$

In particular, $\|\widehat{M}q - \widehat{p}\|_\infty \leq e + \tau = \rho$, so $q \in \widehat{Q}$. The feasible set is closed in the compact simplex. It is nonempty on E , and the continuous linear objective therefore attains its maximum V_ℓ . Using (16),

$$r_\ell^\top q \leq \widehat{r}_\ell^\top q + e \leq V_\ell + e.$$

As $R_\ell(q) \leq 1$, clipping $V_\ell + e$ at one preserves the upper bound on risk. This proves $R_\ell(q) \leq U_\ell$ for every rule on the same event.

Excess certificate. For any $v \in \widehat{Q}$, decompose the true observable difference into three parts:

$$M(v - q) = (M - \widehat{M})v + (\widehat{M}v - \widehat{p}) + (\widehat{p} - Mq).$$

Their infinity norms are at most e, ρ, τ , respectively. Hence $\|M(v - q)\|_\infty \leq 2\rho$, and the definition of the modulus implies $r_\ell^\top v \leq r_\ell^\top q + \Omega_\ell(2\rho)$. The source risk error contributes at most another e :

$$\widehat{r}_\ell^\top v + e \leq r_\ell^\top v + 2e \leq R_\ell(q) + \Omega_\ell(2\rho) + 2e.$$

Taking the maximum over v and clipping at one proves (7). No norm of an inverse or pseudoinverse appears in this argument.

Exact dual value. For this deterministic calculation let $A = \widehat{M}$, $b = \widehat{p}$ and $c = \widehat{r}_\ell$, and assume $\widehat{Q} \neq \emptyset$. For every feasible v and $a \in \mathbb{R}^d$,

$$c^\top v = a^\top Av + (c - A^\top a)^\top v \leq a^\top b + \rho \|a\|_1 + \max_y (c - A^\top a)_y.$$

The inequality uses the coordinatewise residual constraint and the fact that an expectation under v does not exceed the largest coordinate. It establishes weak duality.

For equality, write the primal constraints as $Av \leq b + \rho \mathbf{1}$, $-Av \leq -b + \rho \mathbf{1}$, $\mathbf{1}^\top v = 1$, $v \geq 0$. The finite linear-program dual is

$$\min_{\substack{u, w \geq 0, z \in \mathbb{R} \\ A^\top(u - w) + z \mathbf{1} \geq c}} (b + \rho \mathbf{1})^\top u + (-b + \rho \mathbf{1})^\top w + z.$$

The primal is feasible and its objective is bounded by $\max_y c_y$. The dual is feasible at $u = w = 0$ and $z = \max_y c_y$. Finite-dimensional LP strong duality thus gives equal finite optimal values. These hypotheses permit faces and boundary points; a strictly feasible point is unnecessary.

Set $a = u - w$. The dual objective is $a^\top b + \rho \mathbf{1}^\top(u + w) + z$. For a fixed a , every nonnegative decomposition obeys $u_i + w_i \geq |a_i|$, and equality is achieved by $u = a_+$, $w = a_-$. Since $\rho \geq 0$, that decomposition cannot increase the objective. For these variables the smallest admissible z is $\max_y (c - A^\top a)_y$. Substitution yields exactly (8). The same reduction remains valid at $\rho = 0$.

Residual alignment. Fix a and set $b_a = r_\ell - M^\top a$. For simplex vectors u, v satisfying $\|M(u - v)\|_\infty \leq t$,

$$|r_\ell^\top(u - v)| \leq t \|a\|_1 + |b_a^\top u - b_a^\top v| \leq t \|a\|_1 + \text{osc}(b_a).$$

The last inequality follows because both expectations belong to $[\min_y (b_a)_y, \max_y (b_a)_y]$. Taking the supremum over (u, v) and then the infimum over a proves (9). At $t = 0$ this calculation contains no division or singular limit.

Measurability, selection and endpoints. For fixed sample sizes, the joint vector of all empirical indicator counts has finitely many possible values. The LP value and its empty-set convention are therefore measurable functions of those counts. A least-index selection is measurable, and the assertion also covers any other stipulated measurable selection among certified rules. Simultaneous validity on E gives $R_{\widehat{c}}(q) \leq \alpha$. Independence of the future target point converts that risk into the stated conditional-on-calibration marginal coverage.

The full-label rule has $r_{\text{all}} = 0$, $\widehat{r}_{\text{all}} = 0$ and $\Omega_{\text{all}} = 0$, so assigning its certificate zero preserves every inequality. Off E , an empty polytope is handled by the explicit value one, with the full-label override retained. Positive sample sizes, including one, make every radius finite. If $K = 1$, the simplex is a single point and all preceding bounds remain valid. Thus the common event establishes the claimed certificate, quantitative bound and selection guarantee, while the deterministic arguments establish both representations. $\square$

A.2 PROOF OF PROPOSITION 2

Proof. We prove Proposition 2. We separate the population identification assertion from the statistical interval assertion. The former uses a fixed column-stochastic matrix and a fixed linear risk functional. The latter additionally uses the full observation law (11).

The exact image of the fiber. Fix $p \in M\Delta_K$. The set $F(p) = \{v \geq 0 : \mathbf{1}^\top v = 1, Mv = p\}$ is nonempty by compatibility. Linear equalities and inequalities make it closed and convex, and containment in the finite simplex makes it compact. The function $v \mapsto r^\top v$ is continuous, so it attains a minimum l at v^- and a maximum u at v^+ . If $l < u$, any $z \in [l, u]$ is attained by

$$v_z = (1 - \lambda)v^- + \lambda v^+, \quad \lambda = \frac{z - l}{u - l} \in [0, 1].$$

The vector v_z remains in the fiber and has risk z . If $l = u$, the image is already a singleton. Every fiber member is an admissible target prior with summary law p ; every prior compatible with p belongs to the fiber. Therefore (12) is the exact identified region, rather than only an outer confidence bound.

Row alignment is sufficient. If $r = M^\top a$, then every $v \in F(p)$ satisfies

$$r^\top v = a^\top Mv = a^\top p.$$

The risk image is consequently a singleton for every compatible p . For approximate observable agreement, the same identity and duality of the one and infinity norms give

$$|r^\top (u - v)| = |a^\top M(u - v)| \leq \|a\|_1 \|M(u - v)\|_\infty.$$

Taking the supremum over pairs in (5) and the infimum over representations $M^\top a = r$ proves the claimed modulus bound. Additive constants require no extra span: column stochasticity gives $M^\top \mathbf{1}_d = \mathbf{1}_K$.

Necessity and strict interior priors. Every row of M is orthogonal to $\ker M$. Rank-nullity implies that the row space has dimension $K - \dim(\ker M)$, so it equals $(\ker M)^\perp$. If $r \notin \text{row}(M)$, there is $v \in \ker M$ with $r^\top v \neq 0$. Column stochasticity then gives

$$\mathbf{1}_K^\top v = \mathbf{1}_d^\top Mv = 0.$$

Set $\bar{q} = \mathbf{1}_K/K$ and choose $0 < s < 1/(K\|v\|_\infty)$, which is possible because $v \neq 0$. Define $q_+ = \bar{q} + sv$ and $q_- = \bar{q} - sv$. Every coordinate is at least $1/K - s\|v\|_\infty > 0$, their coordinates sum to one, and

$$Mq_+ = M\bar{q} = Mq_-, \quad |r^\top (q_+ - q_-)| = 2s|r^\top v| > 0.$$

Thus at least one fiber has nonsingleton risk image. This establishes necessity for the all-fiber assertion and provides the required interior pair.

The argument does not imply that every individual fiber has nonzero width. Positivity constraints can remove all ambiguity at a boundary value of p . At $K = 1$, column stochasticity ensures nonzero rank, the kernel is trivial, and every scalar risk belongs to the row space. These cases are consistent with the quantified equivalence.

Uniform interval honesty. Let q_0, q_1 have equal summary distributions and let $\theta_j = r^\top q_j$. Equation (11) makes the complete data law identical at both priors: the target products coincide, and the independent auxiliary factor is the same H . Denote this common law by $\mathbb{P}$. Uniform honesty implies $\mathbb{P}\{\theta_0 \in I\} \geq 1 - \delta$ and $\mathbb{P}\{\theta_1 \in I\} \geq 1 - \delta$. Hence

$$\mathbb{P}\{\theta_0, \theta_1 \in I\} \geq 1 - \mathbb{P}\{\theta_0 \notin I\} - \mathbb{P}\{\theta_1 \notin I\} \geq 1 - 2\delta.$$

An interval containing both endpoints has length at least $|\theta_1 - \theta_0|$. This proves the statistical bound. For $\delta \geq 1/2$, its right-hand side is nonpositive and the inequality remains valid. An unbounded interval also satisfies the length assertion.

Classification realization. In the displayed singular example, take $P_1 = P_2$ to be point mass at feature 1 and P_3 point mass at feature 2. For $C = \{1, 2\}$, the conditional error probabilities are exactly $(0, 0, 1)$, and the risk is $q_3 = p_2$. For $C = \{1\}$ they are $(0, 1, 1)$, and the fiber at $(3/4, 1/4)$ consists of priors $(a, 3/4 - a, 1/4)$ with $0 \leq a \leq 3/4$. Its risk is $1 - a$, whose exact image is $[1/4, 1]$. The two interior priors in the main text give $a = 1/2$ and $a = 1/4$, respectively. They therefore produce risks $1/2$ and $3/4$ while preserving the observation law. These are actual measurable prediction sets under fixed conditional feature laws. Consequently every assertion of Proposition 2 follows. $\square$

A.3 PROOF OF THEOREM 3

Proof. We prove Theorem 3. Fix the full observation experiment (11). Write $P_j = Mq_j$ for the single-summary law and $\theta_j = r^\top q_j$. The proof bounds differences of probabilities under the complete data laws before using uniform interval honesty.

Distances with zero probabilities allowed. For categorical laws P, Q , let $A(P, Q) = \sum_i \sqrt{P_i Q_i} = 1 - h^2(P, Q)/2$. Total variation is half the sum of absolute mass differences. Factoring those differences and applying Cauchy–Schwarz gives

$$\begin{aligned} \text{TV}(P, Q) &\leq \frac{1}{2} \left[\sum_i (\sqrt{P_i} - \sqrt{Q_i})^2 \right]^{1/2} \left[\sum_i (\sqrt{P_i} + \sqrt{Q_i})^2 \right]^{1/2} \\ &= \frac{1}{2} h(P, Q) \sqrt{4 - h^2(P, Q)} \leq h(P, Q). \end{aligned}$$

No coordinate appears in a denominator, so this calculation permits zeros in either probability vector.

Independent products and auxiliary observations. Expanding the finite product sum gives

$$A(P^{\otimes n}, Q^{\otimes n}) = \sum_{i_1, \dots, i_n} \prod_{j=1}^n \sqrt{P_{i_j} Q_{i_j}} = A(P, Q)^n.$$

For $x \in [0, 1]$, $(1 - x)^n \geq 1 - nx$: induction follows by multiplying the bound at n by $1 - x$, which yields $1 - (n + 1)x + nx^2 \geq 1 - (n + 1)x$. Since $h^2(P, Q)/2 \in [0, 1]$, this proves

$$h^2(P^{\otimes n}, Q^{\otimes n}) = 2\{1 - (1 - h^2(P, Q)/2)^n\} \leq nh^2(P, Q).$$

The theorem’s hypothesis therefore implies $\text{TV}(P_0^{\otimes n}, P_1^{\otimes n}) \leq 1/4$.

Appending an independent common-law variable does not alter total variation. Indeed, with counting measure on the finite target strings and measure H on the auxiliary space,

$$\frac{1}{2} \sum_z |P_0^{\otimes n}(z) - P_1^{\otimes n}(z)| dH(w) = \text{TV}(P_0^{\otimes n}, P_1^{\otimes n}).$$

Any randomized algorithm is covered by placing its independent seed in W . Taking a measurable function of these data cannot increase the difference in probabilities of an output event, because that event is the inverse image of a data event. This justifies (14) for the actual interval-producing experiment, not merely for its target marginals.

Transferring coverage. Let $B_j = \{\theta_j \in I\}$. Uniform honesty gives $\mathbb{P}_{q_0}(B_0) \geq 1 - \delta$ and $\mathbb{P}_{q_1}(B_1) \geq 1 - \delta$. The total-variation bound transfers the second event:

$$\mathbb{P}_{q_0}(B_1) \geq \mathbb{P}_{q_1}(B_1) - \frac{1}{4} \geq \frac{3}{4} - \delta.$$

The union bound under $\mathbb{P}_{q_0}$ now gives $\mathbb{P}_{q_0}(B_0 \cap B_1) \geq 3/4 - 2\delta$. On that intersection $|I| \geq |\theta_1 - \theta_0|$, proving (13). Nonnegativity of interval length implies

$$\mathbb{E}_{q_0}|I| \geq (\frac{3}{4} - 2\delta)|\theta_1 - \theta_0|.$$

The coefficient is positive for $\delta \in (0, 1/4)$. Taking the supremum over q on the left and then the supremum over admissible pairs on the right yields the expected-length assertion. An infinite expectation makes the inequality valid without an integrability assumption.

A binary admissible pair. For $0 < \kappa \leq 1/4$, set $t = \min\{1/2, (16\kappa\sqrt{n})^{-1}\}$ and

$$q_0 = (1/2 + t/2, 1/2 - t/2), \quad q_1 = (1/2 - t/2, 1/2 + t/2).$$

Both priors have coordinates in $[1/4, 3/4]$. Their risks for $r = (0, 1)$ differ by t . Multiplication by the displayed matrix gives $P_0 = (1/2 + \kappa t, 1/2 - \kappa t)$ and P_1 with swapped coordinates. Hence

$$h^2(P_0, P_1) = 2 - 2\sqrt{1 - 4\kappa^2 t^2}.$$

For $x = 4\kappa^2 t^2 \in [0, 1]$, $1 - \sqrt{1 - x} = x/(1 + \sqrt{1 - x}) \leq x$. It follows that $h^2(P_0, P_1) \leq 8\kappa^2 t^2$. The definition of t ensures $\kappa t \leq 1/(16\sqrt{n})$ in both branches: if $t = 1/2$, the choice of that branch states precisely this inequality. Therefore

$$h^2(P_0, P_1) \leq \frac{1}{32n} \leq \frac{1}{16n}.$$

The pair is admissible and supplies separation t , proving the claimed lower bound on the supremum.

At $\kappa = 0$, take $t = 1/2$ directly. The columns coincide, the summary distributions coincide, and the same risk separation holds. This checks the endpoint independently of the infinity convention. At $n = 1$ and at $\kappa = 1/4$, all coordinates and inequalities remain in their stipulated ranges; equality of the two branches causes no discontinuity in the construction.

Finally, choose $X = Z \in [2]$ with class-conditional law given by the corresponding column of M_κ . The constant set $\{1\}$ never misses label 1 and always misses label 2, realizing $r = (0, 1)$. The source mechanism is fixed under both target priors, so arbitrarily much independent source information remains a common factor H . Thus the binary pair is a classification problem satisfying the required experiment, and all parts of Theorem 3 are established. $\square$

B EXPERIMENTAL DETAILS

The program uses Python 3.10.20, NumPy 2.2.6, SciPy 1.15.3 and Matplotlib 3.10.9 on an arm64 CPU, without downloaded data or models. There are 500 repetitions per regime. Seed formulas are $71403 + s + 100000n$ for the singular experiment and $3917 + s + 100000n$ for the nonsingular experiment, with $s = 0, \dots, 499$. The weak-identification experiment uses seed $60011 + \lfloor 10^6 \kappa \rfloor$ for a stream of 500 binomial draws.

For the singular model, $M = [e_1, e_1, e_2]$ and the risk vectors are $(0, 0, 1)$ and $(0, 1, 1)$. Source observations are deterministic within each class, but the conservative source radii are retained. The target summary law is $(.75, .25)$; priors $(.5, .25, .25)$ and $(.15, .6, .25)$ give non-aligned risks $.5$ and $.85$. The full-label rule is the third library member. In the nonsingular model, columns of M are $(.8, .2)$ and $(.2, .8)$, and $q = (.15, .85)$. Source counts are multinomial within each stratum. The first rule predicts its summary label, the second always predicts label 1, and the third returns both labels. Error rates and matrix entries share source observations, as permitted by the theorem.

For weak identification, the separation grid is $0, .005, .01, .02, .04, .08, .16, .25$, with $n = 400$ and $q = (.5, .5)$. If B counts summary 1, the equal-tail binomial interval uses endpoints $\text{Beta}^{-1}(\delta/2; B, n - B + 1)$ and $\text{Beta}^{-1}(1 - \delta/2; B + 1, n - B)$, with endpoint zero or one when the respective shape parameter vanishes. For $\kappa > 0$, map it through $q_2 = (.5 + \kappa - p_1)/(2\kappa)$ and intersect with $[0, 1]$. Empty intersections are recorded with length zero; at $\kappa = 0$ the interval is $[0, 1]$.

Linear programs use SciPy's HiGHS backend. The numerical failure test allows 10^{-9} tolerance. Error bars use the empirical standard deviation divided by $\sqrt{500}$ and multiplied by 1.96. Zero-failure upper limits use $1 - .05^{1/500}$. Raw JSON records every repetition; scripts preserve alternative figure designs. Numerical results were unchanged when legends and labels were revised.